Understanding projectiles

AS Level | **9702 AS Ch.2** | Chapter 2: Accelerated motion | Lesson 2.14 | Sections 2.14

Goal: Solve problems for a projectile launched at an angle by resolving its initial velocity.

velocity

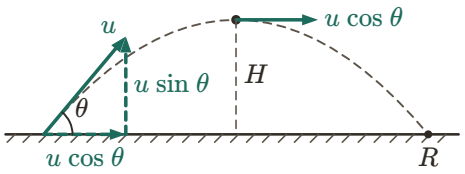
01 The big picture

Resolve the launch velocity into $u\cos\theta$ horizontally (constant all flight) and $u\sin\theta$ vertically (decelerated by g on the way up). At the highest point the vertical velocity is zero, but the horizontal velocity is not.

Over level ground the flight is symmetrical: time up = time down. Without air resistance the range is greatest at 45°. Air resistance lowers the height and the range and makes the descent steeper.

Range [paraphrase]
The horizontal distance travelled before landing.

Fig. 1 · Launched at an angle



At the top only $u\cos\theta$ is left: the vertical velocity is zero.

02 Key equations

Components

$$u_x = u\cos\theta, \quad u_y = u\sin\theta$$

Level ground

$$T = \frac{2u_y}{g}, \quad R = u_x T$$

Max height

$$H = \frac{u_y^2}{2g}$$

Symbols and units

u, θ	launch speed and angle	$\mathrm{m\,s^{-1}}$, °	T	time of flight	s	H, R	max height, range	m

03 Sample problems with solutions

Sample 1 · A kicked ball

A ball is kicked at 20 m s^{-1} at 30° above level ground. Find its time of flight, range and maximum height.

Step 1. Components:

$$u_x = 20\cos 30^\circ = 17.3\ \mathrm{m\,s^{-1}}, \quad u_y = 20\sin 30^\circ = 10.0\ \mathrm{m\,s^{-1}}$$

Step 2. Vertical gives the time; horizontal the range:

$$T = \frac{2\times 10.0}{9.81} = 2.04\ \mathrm{s}, \quad R = 17.3\times 2.04 = 35.3\ \mathrm{m}$$

Step 3. At the top $v_y = 0$:

$$H = \frac{10.0^2}{2\times 9.81} = 5.10\ \mathrm{m}$$

Answer: 2.0 s; 35 m; 5.1 m

Unit check: $\frac{(\mathrm{m\,s^{-1}})^2}{\mathrm{m\,s^{-2}}} = \mathrm{m}$ ✓

04 Practice problems

P1 What is the velocity of a projectile at its highest point?
Answer: $u\cos\theta$, horizontal

P2 Thrown at 25 m s^{-1} at 60°: find the maximum height.
Answer: 23.9 m (24 m)

P3 At 15 m s^{-1} and 40°: find the time of flight.
Answer: 1.97 s (2.0 s)

P4 Which launch angle gives the longest range?
Answer: 45° (no air resistance)

05 Other useful information

Common mistakes

✗ **Wrong:** Use u in the vertical equations.
✓ **Right:** Vertical: $u\sin\theta$. Horizontal: $u\cos\theta$.

Exam tips

- Find T from the vertical motion first; it links the two directions.

What the syllabus asks

- 9702 topic 2: projectile motion without air resistance. Coursebook 2.14.

Explore: Practice questions for this lesson